Multi-LLM Case Study Analysis & Evaluation

Problem → Attempt → Result → Limitation Framework Across Three AI Outputs

Task: Write a clear case study following Problem → Attempt → Result → Limitation structure
Duration: 60 minutes
Evaluation Scope: Perplexity vs. Claude vs. ChatGPT output quality and approach

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Introduction

This document consolidates and evaluates case study outputs from three different LLMs (Perplexity, Claude, and ChatGPT) tasked with demonstrating the Problem → Attempt → Result → Limitation framework. Each model produced a case study, with varying depths of detail, clarity, and supplementary guidance. The evaluation reveals distinct strengths, weaknesses, and applicable contexts for each approach.

Part 1: The Three Case Studies

Case Study 1: E-Commerce Platform Customer Retention (Perplexity)

Domain: E-commerce | **Problem Type:** Business/Operational

Problem
A mid-sized e-commerce platform experienced severe customer churn, losing 25% of users monthly. The root cause: generic, rule-based product recommendations that ignored user behavior data. This manifested as low engagement (2.5-minute average sessions) and severe cart abandonment (70%), directly impacting revenue.

Attempt
The team implemented a collaborative filtering algorithm using Python's Surprise library. Process:

- Trained on six months of historical user interaction data
- A/B tested new system vs. baseline over four weeks
- Integrated via API for real-time personalization

Result
Measurable business impact within the test window:

- Churn: 25% → 15% (40% reduction)
- Session time: 2.5 min → 4.2 min
- Conversion rate: +18% increase

- Revenue impact: +$50,000/month
- User satisfaction: 3.2 → 4.1 out of 5

Limitation

- **Cold start problem:** New users (30% of traffic) had insufficient data for effective recommendations, leaving a significant segment underserved
- **Computational overhead:** Required frequent retraining for seasonal adaptability, creating scaling and cost bottlenecks
- **Unaddressed:** Scalability strategy as user base grows

Case Study 2: Customer Support Response Optimization (Claude)

Domain: SaaS | **Problem Type:** Operational/Staffing

Problem
A mid-sized SaaS company managed 40+ support tickets daily with unacceptable 48-hour response times. 15% of customer churn complaints explicitly cited slow support. Root cause: single support agent managing all channels (email, chat, phone) with no centralized ticketing system.

Attempt
Three-part operational intervention:

1. Implemented Zendesk ticketing system for channel centralization
2. Hired second support agent to distribute workload
3. Established SLAs: 2-hour response for urgent, 24-hour for standard

Timeline: 1 week system setup + 2 weeks training

Result
Four-week outcome metrics:

- Response time: 48 hours → 3.5 hours
- First-contact resolution: 60% → 78%
- Satisfaction scores: 6.8/10 → 8.2/10
- Customer retention: 12 at-risk renewals preserved
- Staffing efficiency: New agent required minimal onboarding

Limitation

- **Symptom vs. root cause:** 30% of tickets originated from missing product documentation—faster response times didn't resolve underlying usability issues
- **Cost structure unsustainability:** Additional agent ($4,500/month) required ongoing justification without deeper fixes
- **Deferred solutions:** Real fix (comprehensive documentation + UX improvements) needed 3 months of engineering time not available in roadmap
- **Fragility:** System dependent on consistent staffing; turnover recreates original bottleneck

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Case Study 3: Evaluating AI Tools as Writing Accelerators (ChatGPT)

Domain: Knowledge Work/Content Production | **Problem Type:** Process Optimization

Problem
Challenge: Converting loosely structured technical thoughts into clear, logically coherent written artifacts (analyses, explanations, documentation). The difficulty wasn't text generation—it was maintaining intent, ensuring logical flow, and avoiding plausible-sounding inaccuracies. Manual writing was slow.

Attempt
ChatGPT used systematically as writing tool:

- Converted raw notes and partial ideas into structured explanations
- Rephrased complex concepts into clearer mental models
- Drafted first-pass versions of long-form content
- Treated outputs as thinking aids, not final authority
- Used explicit prompts about scope, assumptions, and constraints

Result
ChatGPT excelled at:

- Structuring information into readable, logical flows
- Explaining complex systems in plain language
- Reducing "blank page → first draft" time significantly
- Surfacing clearer narrative structures than manual writing
- Providing multiple concept framings for comparison

Limitation

- **Unreliable uncertainty signaling:** Model fills information gaps with plausible extrapolation rather than indicating uncertainty, presented with confident phrasing

- **Unsuitable as authority:** Cannot serve as primary source of truth, technical accuracy, or evidence without external verification
- **Output drift:** Long outputs risk subtle deviation from original constraints
- **Knowledge gaps:** Invents plausible but incorrect details for niche topics or recent events outside training data
- **Implication:** Best used as thinking/articulation accelerator, not judgment replacement

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Part 2: LLM Performance Evaluation

Evaluation Criteria

Criterion	Weight	Description
Clarity	20%	How well structured and easy to follow is the case study?
Specificity	20%	Are metrics, timelines, and details concrete?
Honesty	20%	Does the limitation section genuinely address trade-offs?
Actionability	15%	Can readers learn methodology they could apply?
Self-Awareness	15%	Does the output acknowledge its own constraints?
Domain Fit	10%	Is the example chosen appropriately for the framework?

Perplexity: E-Commerce Case Study

Strengths:

- ✅ Highly specific metrics with clear before/after data
- ✅ Technical depth (mentions specific library: Surprise)
- ✅ Clear timeline (four weeks)
- ✅ Quantified business impact ($50,000/month)
- ✅ Precise limitation identification (30% cold start impact)

Weaknesses:

- ❌ Minimal context on team size, budget, or resource constraints
- ❌ Limited explanation of *why* the approach was chosen

- ❌ No discussion of alternative solutions considered
- ❌ Relatively brief; could benefit from deeper analysis
- ❌ Doesn't explain what "collaborative filtering" means for a non-technical reader

Best For: Technical audiences, data-driven decision makers, ML practitioners
Score: 8.2/10

Claude: Customer Support Case Study

Strengths:

- ✅ Provides template structure alongside example (didactic value)
- ✅ Explicit writing guidance (Problem/Attempt/Result/Limitation tips)
- ✅ Honest limitation analysis (identifies cost unsustainability, unresolved root causes)
- ✅ Accessible to non-technical readers
- ✅ Acknowledges organizational dependencies (staffing fragility)
- ✅ Shows how quick wins can mask deeper issues

Weaknesses:

- ❌ Less technically sophisticated than Perplexity
- ❌ Fewer specific metrics in some areas (doesn't quantify team satisfaction)
- ❌ Template presentation could overshadow case study depth
- ❌ Less discussion of implementation challenges or obstacles

Best For: Managers, process improvement practitioners, organizational learning, teaching case study methodology
Score: 8.5/10

ChatGPT: AI Tool Evaluation Case Study

Strengths:

- ✅ Self-referential (demonstrates the framework by applying it to itself)
- ✅ Exceptional honesty about limitations and constraints
- ✅ Addresses knowledge work (domain relevant to many readers)
- ✅ Clear conclusion with practical guidance

- ✅ Acknowledges nuance (tool is useful in context, not universally)
- ✅ Demonstrates high self-awareness about AI capabilities/gaps

Weaknesses:

- ❌ Less quantified data than Perplexity (no time savings metrics, no ROI calculation)
- ❌ More abstract/subjective than business cases
- ❌ Could benefit from specific examples of "drift" or hallucinations
- ❌ Limitation section is strong but could provide alternative approaches

Best For: AI practitioners, content teams, decision-makers evaluating AI tools, audiences concerned with reliability
Score: 8.3/10

Part 3: Comparative Analysis

Key Differences in Approach

Dimension	Perplexity	Claude	ChatGPT
Case Domain	Technical/Business	Operational	Meta/Knowledge Work
Quantification	Very high (6 metrics)	High (5 metrics)	Low (qualitative focus)
Audience Sophistication	Technical	Generalist	Reflective practitioners
Limitation Depth	Technical constraints	Organizational/systemic	Epistemological
Pedagogical Value	Data-driven	Methodological	Philosophical
Real-world Applicability	Immediate implementation	Immediate implementation	Framework thinking

What Each LLM Revealed About Case Study Quality

Perplexity: Case studies need hard metrics. Numbers build credibility and enable comparison across contexts. However, numbers alone don't tell the full story—context and constraints matter.

Claude: The best case studies teach through honesty about trade-offs. The model understood that limitations aren't failures—they're opportunities to identify what was actually solved vs. deferred.

ChatGPT: Exceptional case studies acknowledge their own epistemic limits. By self-referentially evaluating an AI tool, ChatGPT demonstrated that the framework works when applied honestly, even to uncomfortable truths.

Part 4: Final Result & Synthesis

What Works Across All Three Outputs

1. **Specificity matters more than length.** All three case studies succeeded by providing concrete, measurable details rather than vague claims.
2. **The limitation section is the heart of the analysis.** Generic "limitations exist" doesn't work. Honest assessment of unresolved problems reveals deeper insights:
   - Perplexity: Technical scalability challenges
   - Claude: Organizational sustainability vs. quick fixes
   - ChatGPT: Epistemological unreliability of AI systems
3. **Context determines case study type.** Different domains benefit from different emphasis:
   - Business: Focus on metrics and ROI
   - Organizational: Focus on systemic implications
   - Tool evaluation: Focus on boundary conditions and constraints
4. **The framework is robust.** All three LLMs successfully applied Problem → Attempt → Result → Limitation, suggesting this structure is domain-agnostic and genuinely useful.

Recommendations for Case Study Writing (Synthesis)

Problem Section:

- Quantify the issue (% loss, time impact, cost)
- Show *why* it mattered to the organization
- Identify root cause, not just symptoms
- Specify what previous approaches failed

Attempt Section:

- Describe what was *actually done*, including timeline
- Name specific tools/systems/decisions
- Include resource constraints you faced
- Mention alternatives considered but rejected

Result Section:

- Use metrics (prefer 3+ data points)
- Include both expected and unexpected outcomes
- Specify the timeframe of measurement
- Quantify business/operational impact

Limitation Section (Critical):

- Be honest. This separates learning from propaganda.
- Identify unresolved root causes (what still needs fixing)
- Calculate total cost and ROI, including hidden costs
- Explain why the limitation exists (constraints, priorities, tradeoffs)
- Suggest what should be prioritized next

Part 5: Evaluation Summary

Overall Assessment

Model	Depth	Clarity	Honesty	Technical Detail	Pedagogical Value	Overall
Perplexity	8	9	8	9	7	**8.2/10**
Claude	8	9	9	7	9	**8.5/10**
ChatGPT	8	8	10	6	8	**8.3/10**

When to Use Each Output

Use Perplexity's Approach When:

- Your audience values data and technical specificity
- Quantifiable ROI is central to the narrative
- Implementation details matter more than organizational dynamics

Use Claude's Approach When:

- You're teaching others how to write case studies
- Organizational context and sustainability matter
- You need to highlight systemic issues vs. quick fixes

Use ChatGPT's Approach When:

- Evaluating tools, processes, or methodologies
- Epistemological honesty is important (what do we actually know?)
- The subject requires acknowledging its own limitations
- Self-referential examples strengthen understanding

Conclusion

All three LLMs successfully demonstrated the Problem → Attempt → Result → Limitation framework, but each brought distinct strengths reflecting their design and training. Perplexity excels at data-driven precision; Claude at organizational insight and pedagogy; ChatGPT at philosophical honesty about AI limitations.

The framework itself proved robust across all three applications, suggesting it's a genuinely useful structure for reflective problem-solving and organizational learning. The most mature case studies aren't those that claim perfect solutions—they're those that honestly assess what was solved, what remains unsolved, and what constraints made certain approaches impossible.

Key Takeaway: Effective case studies do three things well: (1) quantify impact with specific metrics, (2) acknowledge organizational/technical constraints honestly, and (3) identify what still needs fixing. The models that did all three strongest were most credible and most useful to readers.